

EDUCATION

School of Mathematical Sciences, University of Science and Technology of China
Hefei, China

Ph.D. in Probability Theory and Mathematical Statistics 09/2019 - 06/2025

- Advisor: Prof. Lijun Bo
- Research area: Mathematical Finance, Stochastic Control and Optimization

School of Mathematical Sciences, University of Science and Technology of China
Hefei, China

B.S. in Mathematics and Applied Mathematics 09/2015- 06/2019

PROFESSIONAL EXPERIENCE

The Hong Kong Polytechnic University

- *Research Assistant* 11/2022 - 09/2025
Department of Applied Mathematics
Supervisor: Prof. Xiang Yu
- *Postdoctoral Fellow* 10/2025 - present
Department of Applied Mathematics
Supervisor: Prof. Xiang Yu

PUBLICATIONS

1. Lijun Bo, Yijie Huang. Dynamic pricing with surging demand. **CSIAM Transactions on Applied Mathematics**, 5(1): 142-181, 2024.
2. Lijun Bo, Yijie Huang, Xiang Yu. A decomposition-homogenization method for Robin boundary problems on the nonnegative orthant. **Electronic Journal of Probability**, 29: 1-25, 2024.
3. Lijun Bo, Yijie Huang, Xiang Yu. Stochastic control problems with state-reflections arising from relaxed benchmark tracking. **Mathematics of Operations Research**, 50(4): 2526-2551. 2025.
4. Lijun Bo, Yijie Huang, Xiang Yu. On optimal tracking portfolio in incomplete markets: The reinforcement learning approach. **SIAM Journal on Control and Optimization**, 63(1): 321-348, 2025.
5. Lijun Bo, Yijie Huang. Optimal inventory control with state dependent jumps. **Advances in Applied Probability**, 57(4): 1360-1391, 2025.
6. Lijun Bo, Yijie Huang, Xiang Yu. An extended Merton problem with relaxed benchmark tracking. **Mathematical Finance**, 36(2): 422-448, 2026.
7. Lijun Bo, Yijie Huang, Kaixin Yan, Xiang Yu. Optimal consumption under relaxed benchmark tracking and consumption drawdown constraint. **SIAM Journal on Financial Mathematics**, 17(1): 78-117, 2026.
8. Lijun Bo, Yijie Huang, Xiang Yu, Tingting Zhang. Continuous-time q-learning for jump-diffusion models under Tsallis entropy. **Mathematics and Financial Economics**, forthcoming, 2026+. (arXiv:2407.03888)

PREPRINTS

1. Lijun Bo, Yijie Huang, Xiang Yu (2025) Mean field game of optimal tracking portfolio. Conditionally accepted as paper with **IEEE Transactions on Automatic Control** (arXiv:2505.01858)
2. Yijie Huang, Mengge Li, Xiang Yu and Zhou Zhou (2025) Continuous-time reinforcement learning for optimal switching over multiple regimes. Major revision with **Finance and Stochastics** (arXiv:2512.04697)
3. Yijie Huang, Kaixin Yan, Qinyi Zhang (2025) Optimal consumption under adjustment costs with respect to multiple reference levels. Submitted. (arXiv:2503.18443)

PREPRINTS	- Xin Guo, Yijie Huang and Xiang Yu (2026) Deterministic policy gradient for learning equilibrium in time-inconsistent control problems. Submitted. (arXiv:2606.11798) - Lijun Bo, Yijie Huang and Tingting Zhang (2026) Optimal consumption and retirement time under shortfall risk measure. Submitted. (arXiv:2606.18935)
WORKING PAPERS	- Yijie Huang, Xiaoqing Liang and Xiang Yu. Optimal annuitization and asset allocation under relaxed benchmark tracking. - Lijun Bo, Yijie Huang and Chenhao Lu. Boundary-aware continuous-time RL for hypercube state spaces via DPG. - Lijun Bo, Dongfang Yang and Yijie Huang. Dynamic Pricing Problem for Platforms in Data Markets.
RESEARCH TALKS AND POSTERS	10th Annual meeting of Financial Engineering and Financial Risk Management Branch, Chengdu, 07/2021 - Recent Advances on Quantitative Finance, The Hong Kong Polytechnic University, 08/2023 - Workshop on Mean Field Models, Control Games, and Related Topics, Tianyuan Mathematical Center in Northwest China, 11/2024 - The 2nd ETH-HK-Imperial Joint Workshop on Quantitative Finance, Hong Kong, 04/2025 - Workshop on Stochastic Control, Financial Technology, and Machine Learning, The Hong Kong Polytechnic University, 12/2025 (Poster) - Symposium on Stochastic Systems, Control, and Games, Xi'an, 05/2026 - The Tenth PKU-NUS Annual International Conference on Quantitative Finance and Economics, Shenzhen, 05/2026
AWARDS AND HONORS	- First-class Academic Scholarship for Ph.D. Students, USTC 2020-2023 - First-class Academic Scholarship for Master Students, USTC 2019
SKILLS	Languages: Chinese, English. Programming: Python, MATLAB, Mathematica.
ACADEMIC SERVICES	Reviewers for: <i>Statistics and Probability Letters, Mathematical Modelling and Analysis</i>